Integrals, Accumulation, and Expected Value

Category: Calculus • Difficulty: Intermediate

Quant Research Field Guide Manual

This volume is designed as a dense single-column field guide for fast review. It explains integration as accumulation, then connects that idea to compounding, discounting, expected value, scenario weighting, option valuation, and practical numeric methods used in quant research.

What this manual emphasizes

Plain-English intuition first, then formulas, market examples, system-fit notes, and implementation ideas.

Who it is for

A serious self-learner building quant math foundations and a reusable research knowledge base.

Visual elements included

Formula blocks, notation tables, worked calculations, comparison tables, process diagrams, synthetic charts, heatmaps, and review sheets.

Single-column vertically stacked reference-card layout • optimized for desktop PDF viewing and mobile reading.

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GUIDE

How to Use This Manual

Quant Research Field Guide

Difficulty: All levels

How to read a concept page

- Start with the one-sentence meaning and memory jogger. Those are the quick-recall anchors.
- Then read the quant-use and market example sections to connect the math to actual research or trading work.
- Use the formula block to see the compact symbolic form, then read the implementation idea to translate the concept into code or workflow.
- Treat the confusion and related-terms sections as review prompts; they are there to stop common misunderstanding.

How to study with it

- On first pass, read sequentially from accumulation to expectation to applications.
- On review passes, jump directly to the cards or formulas you need and use the page titles like an indexed wiki.
- When a page mentions system fit, ask how the concept enters your own data $\rightarrow$ features $\rightarrow$ model $\rightarrow$ signal $\rightarrow$ backtest pipeline.
- If a page feels intuitive but not operational, replicate the implementation idea in a small notebook.

What to focus on most

- Understand what the integrand represents and what units the final integral should have.
- Understand the difference between accumulation over time and expectation over states.
- Understand that weighted averaging, compounding, and discounting are all specialized forms of accumulation logic.
- Understand when to use closed forms, simple sums, numeric integration, or Monte Carlo.

GUIDE

Core Notation and Quick Symbols

Quant Research Field Guide

Difficulty: Intermediate

Notation Table

∫_a^b f(x) dx	Definite integral of f from lower bound a to upper bound b. Usually interpreted as total accumulation across that interval.
F(x)+C	Antiderivative family of f(x); derivative of F returns f.
Δx or Δt	Small step size in x or time used in numerical approximations or sampled data.
E[X]	Expected value of random variable X.
f(x)	Context-dependent notation: a generic function, or often a probability density function in probability pages.
F(x)	Often the cumulative distribution function in probability pages; context matters.
PV	Present value.
W_t	Wealth at time t.
R_t or r_t	Return over a period or instantaneous return/rate, depending on context.
Cov(X,Y)	Covariance between X and Y.

Quick Symbol Reminder

`Accumulation over time: A(t)=∫_0^t g(s)ds | Expectation over states: E[h(X)]=∫ h(x)f(x)dx`

The first integral adds contributions through time; the second adds contributions across possible outcomes weighted by probability.

Interpretation Reminder

- Ask what variable you are integrating over: time, price, state, probability, or something else.
- Ask whether the result should be a running function, a single total number, or an expected value across scenarios.
- Ask whether the integral is exact, numerically approximated, or estimated by simulation.

CALCULUS

Why Integrals Matter in Quant Research

Quant Research Field Guide

Difficulty: Intermediate

One-Sentence Meaning

An integral turns a local rate, density, or stream of increments into a total quantity, which makes it the natural language for accumulation, valuation, and expectation.

Memory Jogger

If derivatives ask “how fast is it changing now?”, integrals ask “how much has built up overall?”

Why It Matters in Quant Research

Cumulative PnL, risk budget usage, discounted value, probability-weighted payoffs, signal evidence, and continuous-time models all depend on accumulation logic.

Simple Market / Trading Example

A strategy earns small intraday returns every minute. The minute-by-minute rate is local information; the integrated sum over the session is the total PnL.

Formula Intuition

```
Total = ∫ rate(t) dt ; Expected payoff = ∫ payoff(x) f(x) dx
```

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

Integrals sit in the aggregation layer of the research stack: raw observations → modeled rates / densities → accumulated quantities → decisions. Implementation idea: In code, most practical integrals are computed as cumulative sums, trapezoid rules, Monte Carlo averages, or rolling aggregations.

Confusions to Avoid + Related Terms

The integral is not “just area from school.” In finance it usually stands for total cash, total probability weight, total exposure, or total expected payoff. Related terms: Riemann sum, definite integral, expected value, continuous compounding, discounted cash flow.

CALCULUS

Riemann Sums

Quant Research Field Guide

Difficulty: Intermediate

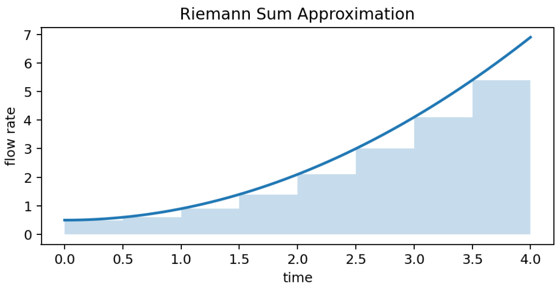
One-Sentence Meaning

A Riemann sum approximates an integral by chopping an interval into small pieces and adding rectangular contributions.

Memory Jogger

Tiny boxes added together become a smooth total.

Visual Block



Visual aid generated for this manual. Use it as intuition support, not as a source of exact numeric values.

Why It Matters in Quant Research

This is the discrete bridge to continuous thinking. Any backtest that adds returns, costs, or exposures over fine time steps is using Riemann-sum logic.

Simple Market / Trading Example

If slippage cost is estimated every 5 seconds, total slippage for the order is approximately the sum of cost_rate_i × Δt.

Formula Intuition

$\int_a^b f(x)\, dx \approx \Sigma f(x_i^*)\, \Delta x$

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

Useful whenever the real object is continuous but the dataset is sampled: prices, order flow, inventory, or volatility processes.
Implementation idea: In Python, np.sum(f_values * delta_x) is the most literal version. With unequal spacing, store each local step width separately.

Confusions to Avoid + Related Terms

The approximation improves when the partitions become finer, but noise or bad interpolation can still produce bad totals. Related terms: Definite integral, numerical integration, trapezoid rule.

CALCULUS

Definite Integrals

Quant Research Field Guide

Difficulty: Intermediate

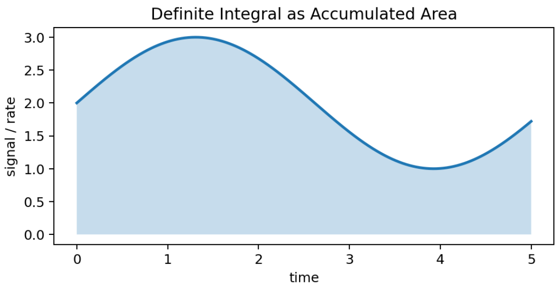
One-Sentence Meaning

A definite integral accumulates the contribution of a function over a specific interval.

Memory Jogger

The lower and upper bounds tell you where accumulation starts and stops.

Visual Block



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Why It Matters in Quant Research

Whenever the question is “how much over this period or region?”, the definite integral is the right object.

Simple Market / Trading Example

Integrating instantaneous volatility proxy over a trading session gives total realized variation exposure over that interval.

Formula Intuition

```
I = ∫_a^b f(x) dx
```

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

Bounded accumulation is used in rolling windows, evaluation horizons, event studies, and finite investment horizons. Implementation idea: Most libraries offer np.trapz or scipy.integrate routines when the curve is known only numerically.

Confusions to Avoid + Related Terms

The result of a definite integral is a number, not another function. Also, the sign matters: contributions below zero subtract from the total. Related terms: Indefinite integral, accumulation function, signed area.

CALCULUS

Indefinite Integrals and Antiderivatives

Quant Research Field Guide

Difficulty: Intermediate

One-Sentence Meaning

An indefinite integral is the family of functions whose derivative equals the original function.

Memory Jogger

Differentiate to check; integrate to reconstruct up to a constant.

Why It Matters in Quant Research

Antiderivatives matter when you want closed-form accumulation, such as converting a rate function into a total value function.

Simple Market / Trading Example

If a toy model says cash-flow rate is 3t^2, then the total accumulated cash flow is t^3 plus a constant determined by the starting condition.

Formula Intuition

```
∫ f(x) dx = F(x) + C where F'(x)=f(x)
```

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

Useful for symbolic understanding, derivations, and sanity checks even if live research mostly uses numeric methods. Implementation idea: Symbolic tools like SymPy are helpful for simple derivations, but production pipelines usually evaluate definite totals numerically.

Confusions to Avoid + Related Terms

Indefinite integrals and definite integrals are related but not the same object. One is a family of functions; the other is a specific evaluated total. Related terms: Fundamental theorem of calculus, constants of integration.

CALLCULUS

The Fundamental Theorem of Calculus

Quant Research Field Guide

Difficulty: Intermediate

One-Sentence Meaning

The fundamental theorem connects differentiation and integration: antiderivatives let us evaluate definite integrals by endpoint subtraction.

Memory Jogger

Differentiate accumulation to recover the local rate; evaluate antiderivatives at the ends to get the total.

Why It Matters in Quant Research

This theorem justifies why a closed-form integral can be turned into a simple difference F(b) – F(a), which is central in tractable models.

Simple Market / Trading Example

A continuous carry model may have an easily integrated rate, allowing total carry over a horizon to be computed exactly.

Formula Intuition

```
If F'(x)=f(x), then ∫_a^b f(x) dx = F(b)-F(a)
```

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

Important when converting between local dynamics and cumulative outcomes in derivations or model notes. Implementation idea: In a notebook, compare a symbolic result to a numeric trapezoid estimate as a validation check.

Confusions to Avoid + Related Terms

The theorem does not say every practical problem has a neat symbolic solution; it says the concepts are linked. Related terms: Antiderivative, accumulation function, endpoint evaluation.

Accumulation Functions

Difficulty: Intermediate

One-Sentence Meaning

An accumulation function $A(t)$ tracks the total built up from a starting point to time t.

Memory Jogger

Think “running total generated by a continuous rate.”

Why It Matters in Quant Research

Cumulative return, cumulative transaction cost, cumulative volume, and cumulative risk exposure are all accumulation functions.

Simple Market / Trading Example

$A(t)=\int_0^t r(s)ds$ maps an instantaneous return or drift rate into accumulated log wealth.

Formula Intuition

$A(t)=\int_0^t f(s)\,ds$ and $A'(t)=f(t)$

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

They are state variables that summarize a whole path with one number at each time. Implementation idea: Use cumulative sums or numerical integration on sampled rates to construct $A(t)$ and plot it beside the original series.

Confusions to Avoid + Related Terms

The derivative of the accumulation function is the local rate, not the total itself. Related terms: Definite integral, cumulative PnL, continuous compounding.

CALCULUS

Area Under the Curve

Quant Research Field Guide

Difficulty: Intermediate

One-Sentence Meaning

The geometric picture of integration interprets the total as the signed area between the curve and the axis.

Memory Jogger

Area is a picture; accumulation is the meaning.

Why It Matters in Quant Research

The picture is useful because it explains why positive contributions add and negative ones offset, which mirrors PnL, flow, and risk contributions.

Simple Market / Trading Example

If daily edge is positive on some days and negative on others, the net edge is the signed area of edge over time.

Formula Intuition

```
Net total = positive area − negative area
```

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

This picture helps during exploratory analysis, chart annotation, and explaining concepts to yourself quickly. Implementation idea: Annotate charts by shading regions above and below zero to make contribution logic visually obvious.

Confusions to Avoid + Related Terms

In many finance settings the “height” is not a literal physical height but a rate, density, or marginal contribution. Related terms: Signed area, net flow, definite integral.

CALCULUS

Signed Area and Net Flow

Quant Research Field Guide

Difficulty: Intermediate

One-Sentence Meaning

Signed area keeps the sign of each contribution so negative regions subtract from the accumulated total.

Memory Jogger

Below-zero mass is not ignored; it reverses direction.

Why It Matters in Quant Research

Critical for net order flow, net carry, cumulative alpha, and inventory changes where opposite-signed contributions cancel.

Simple Market / Trading Example

A market-making book may gain spread at one time and lose from adverse selection later; the net effect is the signed accumulation of both.

Formula Intuition

```
Net = ∫ f(t) dt ; Gross = ∫ |f(t)| dt
```

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

Signed accumulation is central whenever gross and net quantities must be separated. Implementation idea: Track gross and net cumulative curves side by side. Large divergence often signals offsetting behavior or choppy execution.

Confusions to Avoid + Related Terms

Net can be small even when gross activity is large. Always inspect both the gross magnitude and the signed total. Related terms: Gross vs net, area under the curve, PnL attribution.

CALCULUS

Continuous Compounding

Quant Research Field Guide

Difficulty: Intermediate

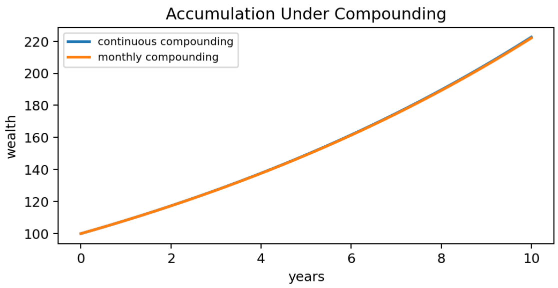
One-Sentence Meaning

Continuous compounding models wealth growth as accumulation of an instantaneous growth rate.

Memory Jogger

When compounding becomes continuous, exponentials appear naturally.

Visual Block



Visual aid generated for this manual. Use it as intuition support, not as a source of exact numeric values.

Why It Matters in Quant Research

Log returns, drift terms, and continuous-time models all use this viewpoint because multiplicative growth becomes additive in logs.

Simple Market / Trading Example

If capital grows at rate r continuously, wealth evolves as W(t)=W0 e^(rt). With time-varying rates, the exponent is an integral.

Formula Intuition

`W(t)=W_0 exp(∫_0^t r(s) ds)`

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

This is a bridge between simple investment math and stochastic calculus thinking. Implementation idea: Convert multiplicative growth to cumulative log returns, then exponentiate back when you need wealth levels.

Confusions to Avoid + Related Terms

Continuous compounding is a modeling convenience, not a claim that trading happens in literally unbroken time. Related terms: Log returns, accumulation function, instantaneous rate.

Integrating Instantaneous Returns

Difficulty: Intermediate

One-Sentence Meaning

Instantaneous returns or drift rates can be integrated over time to obtain accumulated log-return or total growth impact.

Memory Jogger

Local return rate × time, added continuously, becomes total log growth.

Why It Matters in Quant Research

This view is foundational in diffusion models, yield curves, factor carry, and any place a continuously varying return rate is modeled.

Simple Market / Trading Example

If short-rate estimates are available every hour, the cumulative carry over the day can be approximated by integrating those hourly rates.

Formula Intuition

```
log(W_t/W_0) = ∫_0^t r(s) ds
```

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

Lets you move between rate forecasts and horizon forecasts in a disciplined way. Implementation idea: Use cumulative sums of small log returns for stability, especially in long backtests.

Confusions to Avoid + Related Terms

Simple returns add only approximately for small moves; log returns add exactly across time. Related terms: Log returns, compounding, cumulative PnL.

CALCULUS

Discounted Cash Flow as an Integral

Quant Research Field Guide

Difficulty: Intermediate

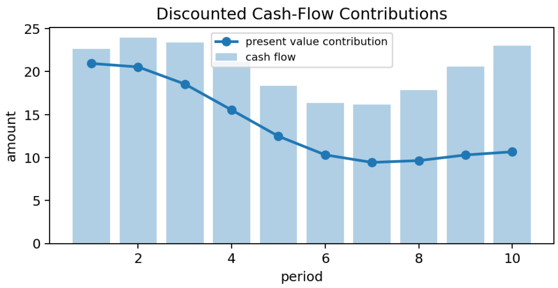
One-Sentence Meaning

Present value is the integral of future cash-flow intensity weighted by a discount factor.

Memory Jogger

Value equals future cash, shrunk by time and discounting, then accumulated.

Visual Block



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Why It Matters in Quant Research

DCF logic shows up in fixed income, project valuation, derivative pricing under risk-neutral measures, and any cash-flow forecasting problem.

Simple Market / Trading Example

A coupon stream can be seen as many small future cash contributions, each discounted back and summed into today’s value.

Formula Intuition

```
PV = ∫_0^T CF(t) · D(t) dt
```

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

Useful whenever your pipeline forecasts a stream rather than a single payoff. Implementation idea: For discrete cash flows, compute pv = sum(cf_t * discount_t). For dense streams, use numerical integration on a modeled curve.

Confusions to Avoid + Related Terms

Discounting is not the same as compounding. One moves value backward to today; the other moves value forward into the future. Related terms: Present value, yield curve, expected payoff.

CALCULUS

Cumulative PnL Through Time

Quant Research Field Guide

Difficulty: Intermediate

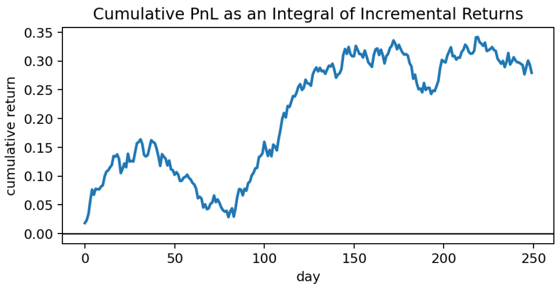
One-Sentence Meaning

Cumulative PnL is the time integral or sum of incremental gains and losses.

Memory Jogger

The equity curve is the accumulated history of tiny wins and losses.

Visual Block



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Why It Matters in Quant Research

Backtests are accumulation machines. All performance metrics begin with a correctly accumulated return or PnL path.

Simple Market / Trading Example

A strategy with noisy daily returns can still show a clear positive cumulative trajectory if the average increment is positive.

Formula Intuition

$PnL(T)=\int_0^T dPnL(t)$ or $\Sigma_t\, pnl_t$

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

This page links calculus directly to the core backtesting object: the equity curve. Implementation idea: Store both incremental pnl and cumulative pnl. Diagnostics on the increment series explain the shape of the curve.

Confusions to Avoid + Related Terms

A smooth cumulative curve can hide path risk, leverage spikes, or dependence on a few large days. Related terms: Running sum, drawdown, log returns, path dependence.

CALCULUS

Execution Cost Accumulation

Quant Research Field Guide

Difficulty: Intermediate

One-Sentence Meaning

Execution cost is often modeled as a rate or local cost contribution that accumulates over the life of an order.

Memory Jogger

Every slice pays a little; the whole trade pays the total.

Why It Matters in Quant Research

Slippage, impact, borrow, and commission are rarely one-shot numbers in realistic execution models.

Simple Market / Trading Example

If impact cost rises with participation rate, the total order cost is the integral of instantaneous impact over the execution horizon.

Formula Intuition

$$\text{Total cost} = \int c(q_t, v_t, t)\, dt$$

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

Critical for turning a paper signal into a realistic strategy by moving from gross alpha to net alpha. Implementation idea: Accumulate commissions, spread, and model-based impact separately before summing them into net execution cost.

Confusions to Avoid + Related Terms

Average cost per share is a summary. The underlying object is usually a path of costs through time or quantity. Related terms: Riemann sums, control variable, optimal execution.

CALCULUS

Exposure and Risk Budget Accumulation

Quant Research Field Guide

Difficulty: Intermediate

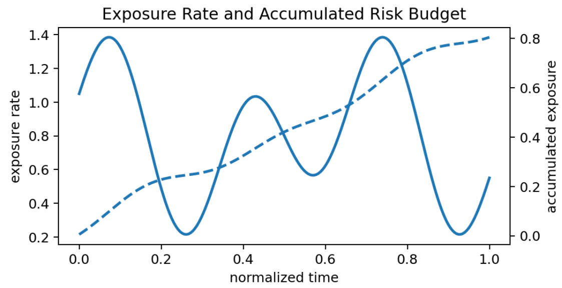
One-Sentence Meaning

Risk usage can be viewed as the accumulation of instantaneous exposure over time.

Memory Jogger

Exposure is the rate; budget consumption is the total.

Visual Block



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Why It Matters in Quant Research

Helps explain time-under-risk, inventory burden, overnight exposure, and how long a portfolio stays loaded.

Simple Market / Trading Example

Two strategies with the same average leverage can have very different accumulated exposure if one stays levered much longer.

Formula Intuition

```
Budget usage ≈ ∫ |exposure(t)| dt
```

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

Useful in portfolio controls, compliance summaries, and diagnostic reporting. Implementation idea: Integrate absolute exposure or VaR estimates over time to see how aggressively risk was actually consumed.

Confusions to Avoid + Related Terms

Peak exposure and cumulative exposure are different. One measures height; the other measures total duration-weighted burden. Related terms: Risk budgeting, time under water, integral of leverage.

PROBABILITY LINK

Probability Weights Intuition

Quant Research Field Guide

Difficulty: Intermediate

One-Sentence Meaning

Expected value works by weighting each possible outcome by how much probability mass it carries.

Memory Jogger

Important outcomes are not the biggest ones; they are the ones with big payoff times meaningful probability.

Why It Matters in Quant Research

This is the basic valuation logic behind expected payoff, scenario analysis, and model averaging.

Simple Market / Trading Example

A rare huge win may still contribute less to expectation than a modest common outcome if its probability is tiny.

Formula Intuition

$E[X] = \Sigma x_i p_i$ or $\int x f(x) dx$

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

A good bridge page between calculus language and probability language. Implementation idea: When you create scenario tables, include both payoff and probability columns so contribution = payoff × probability is explicit.

Confusions to Avoid + Related Terms

Probability is a weight, not a guarantee. A positive expected value does not mean a positive next outcome. Related terms: Expected value, probability density, scenario analysis.

PROBABILITY LINK

Expected Value for Discrete Outcomes

Quant Research Field Guide

Difficulty: Intermediate

One-Sentence Meaning

The expected value of a discrete variable is the probability-weighted average across its possible values.

Memory Jogger

List the states, weight them, and add them.

Why It Matters in Quant Research

Useful for scenario trees, event trades, earnings bets, and finite-state stress tests.

Simple Market / Trading Example

If a trade makes +3 with probability 0.6 and loses 2 with probability 0.4, its expected value is 1.0.

Formula Intuition

`E[X]=Σ_i x_i p_i`

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

This is often the first expectation object in a research workflow because it is easy to audit. Implementation idea: Represent scenarios as a DataFrame with columns scenario, payoff, probability, and contribution.

Confusions to Avoid + Related Terms

The expectation need not be an outcome that ever occurs; it is a center of mass, not a promised realized value. Related terms: Scenario analysis, weighted average, expectation contribution.

PROBABILITY LINK

Expected Value for Continuous Outcomes

Quant Research Field Guide

Difficulty: Intermediate

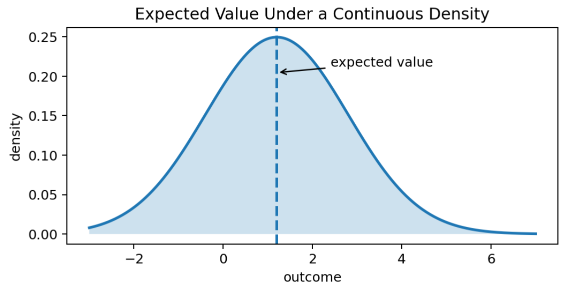
One-Sentence Meaning

For a continuous variable, expectation is an integral of outcome value times probability density.

Memory Jogger

A density spreads probability continuously across the line, so sums become integrals.

Visual Block



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Why It Matters in Quant Research

Option pricing, continuous return models, and risk forecasting often live in continuous state spaces.

Simple Market / Trading Example

The expected terminal price under a model is found by integrating price × density over all possible prices.

Formula Intuition

$$E[X]=\int_{-\infty}^{\infty} x\, f(x)\, dx$$

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

This lets you move from scenario tables to model-based densities when the state space is large or continuous. Implementation idea: For sampled densities, multiply each grid value by the density and grid spacing, then sum.

Confusions to Avoid + Related Terms

Density is not probability itself. Probability comes from integrating density over a region. Related terms: Probability density function, CDF, expected payoff.

PROBABILITY LINK

Probability Density Functions

Quant Research Field Guide

Difficulty: Intermediate

One-Sentence Meaning

A probability density function describes how probability mass is distributed across continuous outcomes.

Memory Jogger

The density can be tall without giving more than 100% probability because only area matters.

Why It Matters in Quant Research

Densities are used in distributional forecasts, option-implied views, and likelihood-based modeling.

Simple Market / Trading Example

A narrow density around a forecast indicates concentration; a wide density indicates uncertainty.

Formula Intuition

$f(x) \geq 0$ and $\int f(x)\, dx = 1$

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

Densities are richer than point forecasts because they support expected values, tail probabilities, and scenario comparisons.
Implementation idea: Normalize numeric density estimates so the total integrated mass equals one before using them for valuation or expectation.

Confusions to Avoid + Related Terms

High density at one point does not mean high probability at that exact point; exact-point probability in a continuous model is zero.
Related terms: CDF, expectation, variance, likelihood.

PROBABILITY LINK

Cumulative Distribution Functions and Tails

Quant Research Field Guide

Difficulty: Intermediate

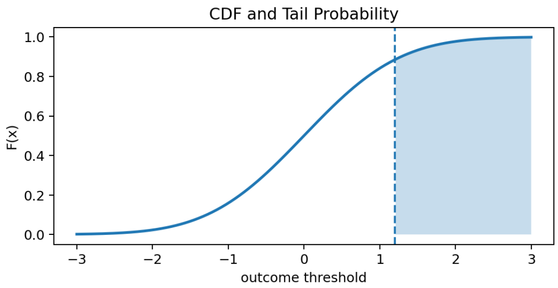
One-Sentence Meaning

The cumulative distribution function gives the probability that a variable is less than or equal to a threshold.

Memory Jogger

CDF answers threshold questions; tail probability answers exceedance questions.

Visual Block



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Why It Matters in Quant Research

Risk limits, stop probabilities, default probabilities, and event exceedance logic are all CDF problems.

Simple Market / Trading Example

To estimate the chance a loss exceeds 5%, compute the upper-tail probability 1 – F(–5%).

Formula Intuition

```
F(x)=P(X≤x) = ∫_{-∞}^{x} f(u) du
```

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

This is the threshold layer of the research system: not just what is expected, but what is likely to breach a rule. Implementation idea: Use empirical CDFs or parametric CDFs to compare forecasted tail risk against realized thresholds.

Confusions to Avoid + Related Terms

The CDF accumulates probability, while the density gives local concentration. Do not treat them as interchangeable plots. Related terms: Density, tail risk, quantiles.

PROBABILITY LINK

Linearity of Expectation

Quant Research Field Guide

Difficulty: Intermediate

One-Sentence Meaning

Expectation is linear, so the expected value of a sum is the sum of expected values.

Memory Jogger

Expectations add cleanly even when the realized path does not.

Why It Matters in Quant Research

Portfolio expectation, component PnL expectation, and aggregated feature contributions all use this property.

Simple Market / Trading Example

Expected portfolio return equals the sum of weighted expected component returns.

Formula Intuition

```
E[aX + bY] = aE[X] + bE[Y]
```

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

This property makes decomposition and attribution much easier in complex systems. Implementation idea: Compute expected contributions by component first; then sum them for a transparent total.

Confusions to Avoid + Related Terms

Linearity of expectation does not imply variance is linear. Risk aggregation is harder than mean aggregation. Related terms: Portfolio return, covariance, attribution.

PROBABILITY LINK

Conditional Expectation

Quant Research Field Guide

Difficulty: Intermediate

One-Sentence Meaning

Conditional expectation is the expected value of a variable given some information set or conditioning event.

Memory Jogger

Expectation changes when you know more.

Why It Matters in Quant Research

All predictive modeling is implicitly conditional: E[return | features], E[loss | regime], E[fill rate | queue state].

Simple Market / Trading Example

A mean-reversion trade may have different expected payoff in quiet versus volatile regimes.

Formula Intuition

```
m(x)=E[Y | X=x]
```

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

Conditional expectation is the core mapping from features to forecasts. Implementation idea: Bucket data by state or fit a model to estimate conditional means, then test stability across subsamples.

Confusions to Avoid + Related Terms

Conditional expectation is not necessarily a single event outcome; it is an average within the conditioned subset. Related terms: Regression, state dependence, law of iterated expectations.

PROBABILITY LINK

Law of Iterated Expectations

Quant Research Field Guide

Difficulty: Intermediate

One-Sentence Meaning

The unconditional expectation equals the expectation of a conditional expectation.

Memory Jogger

Average the state-specific averages and you recover the overall average.

Why It Matters in Quant Research

Lets you separate a problem into regimes, forecast within each regime, and then combine them coherently.

Simple Market / Trading Example

Overall expected return can be written as the probability-weighted average of expected return in bull, bear, and neutral states.

Formula Intuition

```
E[Y] = E[E[Y | X]]
```

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

Very useful in hierarchical models and scenario pipelines. Implementation idea: Build regime-conditioned estimates first, then aggregate using current or historical regime probabilities.

Confusions to Avoid + Related Terms

If the regime probabilities or regime-level expectations are wrong, the final unconditional expectation is wrong too. Related terms: Conditional expectation, scenario analysis, mixture models.

PROBABILITY LINK

Expectation and Variance

Quant Research Field Guide

Difficulty: Intermediate

One-Sentence Meaning

Expectation tells you the center of a distribution, while variance tells you how dispersed outcomes are around that center.

Memory Jogger

Mean is where outcomes balance; variance is how much they spread.

Why It Matters in Quant Research

You need both reward and uncertainty to evaluate strategies sensibly.

Simple Market / Trading Example

Two trades may have the same expected value but radically different risk because one has much wider outcome dispersion.

Formula Intuition

```
Var(X)=E[(X-E[X])^2]=E[X^2]-(E[X])^2
```

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

This is the minimum pair needed for many first-pass portfolio and strategy diagnostics. Implementation idea: Always compute mean and standard deviation together before interpreting a forecast or backtest result.

Confusions to Avoid + Related Terms

High expectation is not enough if variance is huge or the path contains unacceptable tail risk. Related terms: Standard deviation, Sharpe ratio, expected value.

PROBABILITY LINK

Covariance and Expectation of Products

Quant Research Field Guide

Difficulty: Intermediate

One-Sentence Meaning

Covariance measures whether two variables tend to move together relative to their means.

Memory Jogger

It is about co-movement, not just individual spread.

Why It Matters in Quant Research

Portfolio construction and factor modeling depend on expectations of cross-products, not just separate averages.

Simple Market / Trading Example

A hedge can reduce risk if the hedge PnL tends to rise when the main strategy loses, producing negative covariance.

Formula Intuition

$Cov(X,Y)=E[(X-E[X])(Y-E[Y])] = E[XY]-E[X]E[Y]$

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

Essential for moving from single-trade expectation to portfolio-level expectation and risk. Implementation idea: Estimate covariance matrices carefully; noisy covariance estimates can dominate portfolio errors.

Confusions to Avoid + Related Terms

Zero covariance means no linear co-movement, not necessarily full independence. Related terms: Correlation, variance, portfolio risk.

MARKET APPLICATION

Expected Payoff of an Option

Quant Research Field Guide

Difficulty: Intermediate

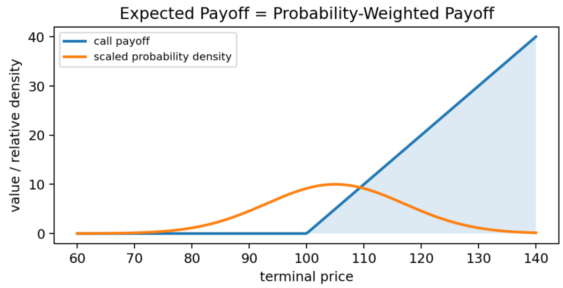
One-Sentence Meaning

An option's value can be understood as the probability-weighted average of its future payoff, usually adjusted for discounting and the chosen pricing measure.

Memory Jogger

Pricing is not about the prettiest payoff line; it is about weighting that line by plausible outcomes.

Visual Block



Visual aid generated for this manual. Use it as intuition support, not as a source of exact numeric values.

Why It Matters in Quant Research

This is the cleanest practical example of expected value as an integral in markets.

Simple Market / Trading Example

A call option has zero payoff below strike and rising payoff above strike. Expected value depends heavily on how much probability mass sits in the right tail.

Formula Intuition

```
Price = e^{-rT} ∫ payoff(S_T) f_Q(S_T) dS_T
```

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

Connects calculus, probability, and derivatives into one concrete object. Implementation idea: On a grid of terminal prices, multiply payoff by density and spacing, sum, then discount.

Confusions to Avoid + Related Terms

Real-world expectation and risk-neutral expectation are not the same thing; pricing uses a specific measure. Related terms: Payoff function, density, discounting.

MARKET APPLICATION

Expected Shortfall Intuition

Quant Research Field Guide

Difficulty: Intermediate

One-Sentence Meaning

Expected shortfall measures the average loss inside the worst tail beyond a chosen quantile.

Memory Jogger

VaR marks the tail cutoff; expected shortfall looks inside the tail.

Why It Matters in Quant Research

A better tail-risk summary than plain VaR because it accounts for how bad losses are once the threshold is breached.

Simple Market / Trading Example

Two strategies with the same 95% VaR can have very different expected shortfall if one has much fatter extreme-loss tails.

Formula Intuition

```
ES_α = E[L | L ≥ VaR_α]
```

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

Useful for risk dashboards and portfolio controls where tail severity matters. Implementation idea: Compute empirical tail means after sorting losses, or estimate from a fitted distribution if justified.

Confusions to Avoid + Related Terms

Expected shortfall is a conditional expectation over tail outcomes, not just another percentile. Related terms: Tail probability, CDF, conditional expectation.

NUMERICAL METHODS

Monte Carlo Estimation of Expectations

Quant Research Field Guide

Difficulty: Intermediate

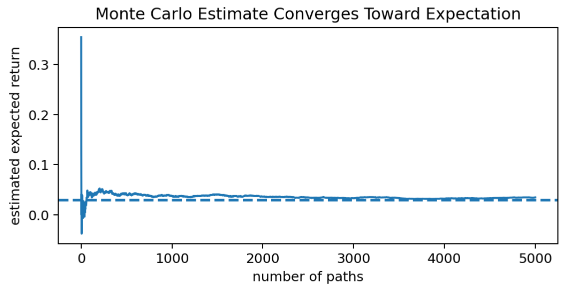
One-Sentence Meaning

Monte Carlo estimates an expectation by simulating many scenarios and averaging the resulting payoffs.

Memory Jogger

If integration is hard analytically, sample the world many times and average.

Visual Block



Visual aid generated for this manual. Use it as intuition support, not as a source of exact numeric values.

Why It Matters in Quant Research

Useful for path-dependent payoffs, nonlinear portfolio risk, and high-dimensional problems where grid integration is impractical.

Simple Market / Trading Example

To estimate an option price under a stochastic model, simulate many terminal prices or paths, evaluate payoff on each, and average.

Formula Intuition

```
E[g(X)] ≈ (1/N) Σ g(X_i)
```

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

This is the workhorse bridge between theoretical expectation and practical estimation. Implementation idea: Track running estimates and confidence intervals so you can see whether simulation noise is still too large.

Confusions to Avoid + Related Terms

More simulations reduce sampling error slowly, at a rate proportional to 1 over the square root of sample size. Related terms: Law of large numbers, simulation error, numerical integration.

NUMERICAL METHODS

Trapezoid Rule

Quant Research Field Guide

Difficulty: Intermediate

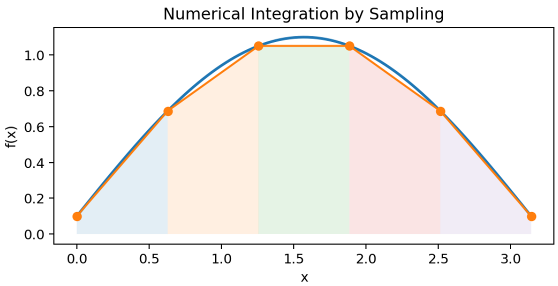
One-Sentence Meaning

The trapezoid rule approximates an integral by connecting sample points with straight lines and summing trapezoid areas.

Memory Jogger

Better than pure rectangles when the curve is reasonably smooth.

Visual Block



Visual aid generated for this manual. Use it as intuition support, not as a source of exact numeric values.

Why It Matters in Quant Research

A robust default when you have ordered grid data such as yields, hazard curves, density estimates, or time-series rates.

Simple Market / Trading Example

Integrating a sampled term-structure curve to get an accumulated discount exponent is naturally a trapezoid-rule problem.

Formula Intuition

$\int_a^b f(x)dx \approx \Sigma ((f_i + f_{i+1})/2) \Delta x_i$

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

One of the safest basic numeric integration tools in the research stack. Implementation idea: np.trapz(y, x) or scipy.integrate.trapezoid(y, x) gives a fast and readable implementation.

Confusions to Avoid + Related Terms

If the sampling grid is too coarse or the function is very curved, error can still be material. Related terms: Riemann sums, Simpson’s rule, quadrature.

NUMERICAL METHODS

Simpson’s Rule and Adaptive Integration

Quant Research Field Guide

Difficulty: Intermediate

One-Sentence Meaning

Simpson’s rule uses local parabolic fits to improve integration accuracy, while adaptive methods allocate more evaluation points where the function is difficult.

Memory Jogger

Curvature deserves smarter sampling.

Why It Matters in Quant Research

Helpful when payoff curves or densities are smooth but curved, or when tail behavior deserves extra resolution.

Simple Market / Trading Example

A sharply curved option payoff under a skewed density may need more accurate quadrature than a crude fixed-step rule.

Formula Intuition

```
Simpson on [a,b]: (Δx/3)[f_0 + 4f_1 + 2f_2 + ... + f_n]
```

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

A precision upgrade when a simple trapezoid estimate is not stable enough. Implementation idea: Use SciPy quadrature functions for smooth mathematical functions and sanity-check against a refined trapezoid estimate.

Confusions to Avoid + Related Terms

Higher-order methods are not magic; noisy data can defeat fancy formulas. Related terms: Trapezoid rule, quadrature, interpolation.

CALCULUS TECHNIQUE

Change of Variables

Quant Research Field Guide

Difficulty: Intermediate

One-Sentence Meaning

Change of variables rewrites an integral in a new coordinate system that is easier to interpret or compute.

Memory Jogger

Choose the variable that makes the structure simpler.

Why It Matters in Quant Research

Log-price transformations, standardizing normal variables, and switching between yield and discount representations all use substitution logic.

Simple Market / Trading Example

A lognormal price model becomes easier to analyze by changing variables from price to standardized normal z-scores.

Formula Intuition

$\int f(g(u))\, g'(u)\, du = \int f(x)\, dx$ with $x=g(u)$

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

Useful in derivations, density transformations, and when comparing models defined in different coordinates. Implementation idea: When transforming simulated variables, verify that the transformed density still integrates to one.

Confusions to Avoid + Related Terms

Do not forget the Jacobian term; changing variables changes the scale of the integration measure. Related terms: Density transformation, standardization, integration technique.

CALCULUS TECHNIQUE

Integration by Parts Intuition

Quant Research Field Guide

Difficulty: Intermediate

One-Sentence Meaning

Integration by parts rearranges an integral of a product into a form that may be easier to interpret or evaluate.

Memory Jogger

It is the integral cousin of the product rule.

Why It Matters in Quant Research

Appears in derivations involving weighted moments, discounting times payoff, and continuous-time identities.

Simple Market / Trading Example

Expected value identities and moment formulas often become cleaner after moving a derivative from one factor to another.

Formula Intuition

$$\int u \, dv = uv - \int v \, du$$

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

Mainly useful for theory notes and analytical derivations rather than routine backtesting. Implementation idea: Use symbolic algebra tools to check manipulations when you are learning the technique.

Confusions to Avoid + Related Terms

Pick u and dv strategically. A bad choice only makes the expression uglier. Related terms: Product rule, antiderivative, derivation techniques.

MULTIVARIATE

Double Integrals and Joint Expectations

Quant Research Field Guide

Difficulty: Intermediate

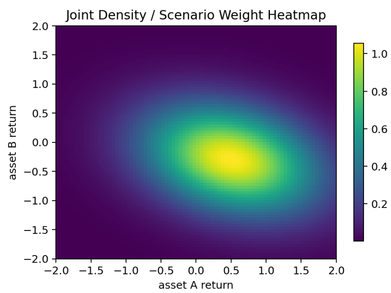
One-Sentence Meaning

A double integral accumulates contributions over a two-dimensional region and is the natural form for expectations under joint distributions.

Memory Jogger

One variable is not enough when payoffs depend on several moving parts.

Visual Block



Visual aid generated for this manual. Use it as intuition support, not as a source of exact numeric values.

Why It Matters in Quant Research

Portfolio payoffs, spread trades, basket options, and multivariate scenario models all require joint reasoning.

Simple Market / Trading Example

Expected payoff of a spread option depends on the joint distribution of both assets, not just their separate means.

Formula Intuition

`E[g(X,Y)] = ■ g(x,y) f(x,y) dx dy`

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

This page expands the manual from one-dimensional intuition to real portfolio settings. Implementation idea: Use scenario grids or Monte Carlo simulation when analytic double integration is not feasible.

Confusions to Avoid + Related Terms

Marginal expectations are not enough when dependence matters. Related terms: Covariance, correlation, joint density.

MARKET APPLICATION

Scenario-Weighted Portfolio Expectation

Quant Research Field Guide

Difficulty: Intermediate

One-Sentence Meaning

Portfolio expected return can be built by assigning each scenario a probability and computing the weighted average of portfolio outcomes.

Memory Jogger

Scenarios are discrete integrals over the state space.

Why It Matters in Quant Research

A practical approximation to continuous expectation and a friendly way to explain model assumptions.

Simple Market / Trading Example

Assign macro regimes probabilities, compute the portfolio return in each regime, then average them.

Formula Intuition

```
E[R_p] = Σ_s P(s) R_p(s)
```

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

Excellent for communicating research logic to yourself or teammates without hiding assumptions. Implementation idea: Maintain a scenario table with regime probabilities, factor shocks, instrument returns, and portfolio contributions.

Confusions to Avoid + Related Terms

Scenario probabilities must sum to one and should be stress-tested because the output is only as good as the state design. Related terms: Discrete expectation, law of iterated expectations, stress testing.

MARKET APPLICATION

Expected Log Growth and Kelly Logic

Quant Research Field Guide

Difficulty: Intermediate

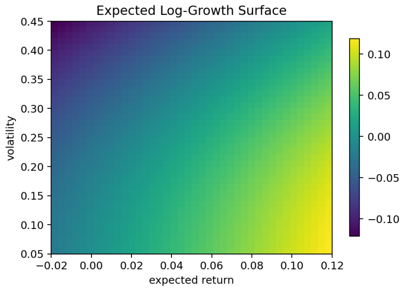
One-Sentence Meaning

Expected log growth focuses on the long-run growth rate of wealth rather than one-period expected wealth.

Memory Jogger

Growing big on average is not the same as growing safely through time.

Visual Block



Visual aid generated for this manual. Use it as intuition support, not as a source of exact numeric values.

Why It Matters in Quant Research

A strategy can have positive expected return yet poor long-run growth if volatility and drawdowns are too large.

Simple Market / Trading Example

Leverage that maximizes expected terminal wealth in one period can still destroy long-run compounded growth.

Formula Intuition

```
Growth objective: maximize E[log(1+fR)]
```

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

This page links expectation to position sizing and survival. Implementation idea: Sweep leverage f across a grid, estimate expected log growth, and inspect how sensitive the optimum is to estimation error.

Confusions to Avoid + Related Terms

Expected value of wealth and expected log wealth answer different questions. Related terms: Compounding, time average, Kelly criterion.

MARKET APPLICATION

Time Average vs Ensemble Average

Quant Research Field Guide

Difficulty: Intermediate

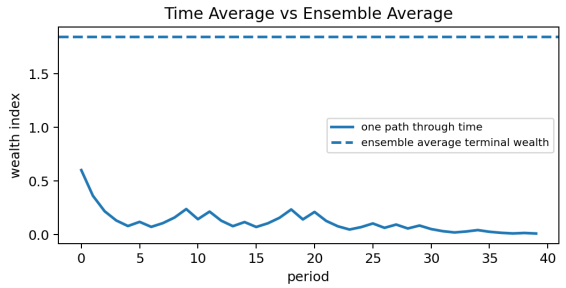
One-Sentence Meaning

The ensemble average looks across many parallel worlds at one horizon, while the time average follows one path through repeated compounding.

Memory Jogger

Investors live through time, not across infinite clones.

Visual Block



Visual aid generated for this manual. Use it as intuition support, not as a source of exact numeric values.

Why It Matters in Quant Research

Important for understanding why volatile positive-expectation strategies can still disappoint in practice.

Simple Market / Trading Example

A gamble may have attractive average terminal wealth across many scenarios but poor typical compounded growth along a single repeated path.

Formula Intuition

`Time-average growth ≈ E[log(1+R)] ; Ensemble mean looks at E[W_T]`

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

This clarifies why path dependence and compounding matter in strategy sizing. Implementation idea: Compare arithmetic mean return, geometric mean return, and expected log growth in the same report.

Confusions to Avoid + Related Terms

Do not evaluate repeated investment decisions using only one-period expected payoff. Related terms: Expected log growth, compounding, path dependence.

INTERPRETATION

Expected Value vs Most Likely Outcome

Quant Research Field Guide

Difficulty: Intermediate

One-Sentence Meaning

The expected value is the average center of mass, while the most likely outcome is the mode.

Memory Jogger

Average and most likely are different in skewed worlds.

Why It Matters in Quant Research

In skewed trades such as options or venture-like bets, the expectation may sit far away from the most probable realized outcome.

Simple Market / Trading Example

A right-skewed trade may lose small amounts most of the time yet still have positive expectation because rare large wins dominate the average.

Formula Intuition

`Mode = argmax f(x), while expectation = ∫ x f(x) dx`

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

Prevents misreading distributions when evaluating strategy behavior. Implementation idea: Plot histograms with both the mode region and mean marked to show the distinction visually.

Confusions to Avoid + Related Terms

Traders often confuse “usually wins” with “has positive expected value.” These are not equivalent. Related terms: Mode, skewness, expected payoff.

PITFALLS

Quant Research Field Guide

Common Mistakes in Accumulation Problems

Difficulty: Intermediate

One-Sentence Meaning

Accumulation problems often fail because the local quantity, the sign convention, the units, or the bounds are defined incorrectly.

Memory Jogger

Most integration mistakes are setup mistakes.

Why It Matters in Quant Research

Bad units or wrong integration bounds can silently corrupt PnL, risk, and valuation calculations.

Simple Market / Trading Example

Integrating annualized volatility as if it were a rate of dollars per day produces nonsense because the units do not match the intended total.

Formula Intuition

```
Check: (units of integrand) × (units of variable) = units of total
```

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

A troubleshooting page for debugging research formulas. Implementation idea: Write units in comments or variable names when prototyping numeric integration code.

Confusions to Avoid + Related Terms

A clean formula does not save a bad definition. Always ask what the integrand means and what units the output should have. Related terms: Dimensional analysis, signed area, numerical integration.

PITFALLS

Quant Research Field Guide

Common Mistakes in Expectation Problems

Difficulty: Intermediate

One-Sentence Meaning

Expectation problems often fail because probabilities are misnormalized, tails are ignored, or the wrong measure is used.

Memory Jogger

An expectation is only as good as its weighting scheme.

Why It Matters in Quant Research

These errors show up in scenario models, derivative pricing, and backtests that confuse sample means with stable expectations.

Simple Market / Trading Example

A payoff grid that forgets to renormalize a density after truncation will understate or overstate value.

Formula Intuition

Sanity check: $\Sigma p_i = 1$ or $\int f(x)\, dx = 1$

Read the formula as the compact rule connecting local contributions to a total quantity or weighted average.

Research-System Fit + Implementation

A troubleshooting page for forecast and valuation logic. Implementation idea: Always run normalization checks and compare empirical estimates to model-implied estimates.

Confusions to Avoid + Related Terms

Do not confuse expected value with median, most likely outcome, or guaranteed result. Related terms: Probability density, normalization, conditional expectation.

Worked Calculation — Discrete Expected PnL

Difficulty: Intermediate

Setup

Suppose a trade has three scenarios: +4 with probability 0.25, +1 with probability 0.50, and –3 with probability 0.25.

Step 1

Compute contributions: 4×0.25 = 1.00 ; 1×0.50 = 0.50 ; (–3)×0.25 = –0.75.

Step 2

Add them: 1.00 + 0.50 – 0.75 = 0.75. The trade has positive expected value of 0.75 units.

Interpretation

Positive expectation does not mean the trade wins most of the time. Here the positive expectation is partly driven by the large upside scenario.

System note

This is the minimum structure for any scenario-based trade note: define states, assign probabilities, compute weighted contributions, and then stress-test the assumptions.

REVIEW / APPENDIX

Worked Calculation — Continuous Expected Payoff

Difficulty: Intermediate

Setup

Suppose a payoff depends on terminal price S and you have a normalized density f(S) defined on a grid. You want E[payoff(S)].

Step 1

Evaluate payoff on each grid point, for example max(S–K,0) for a call option.

Step 2

Multiply pointwise: $\text{contribution}_i = \text{payoff}_i \times \text{density}_i \times \Delta S$.

Step 3

Sum all contributions. The result approximates the integral of payoff times density.

Interpretation

Large payoffs in low-probability states matter only if payoff × probability weight stays meaningful.

Implementation note

Always verify that $\Sigma\, \text{density}_i \times \Delta S \approx 1$ before trusting the expected payoff.

Worked Calculation — Discounting a Cash-Flow Stream

Difficulty: Intermediate

Setup

Assume four annual cash flows: 10, 12, 12, and 15. Use discount rate 8%.

Step 1

Compute present-value contributions: 10/1.08, 12/1.08^2, 12/1.08^3, 15/1.08^4.

Step 2

Numerically these are about 9.26, 10.29, 9.53, and 11.03.

Step 3

Add them for total present value ≈ 40.11.

Interpretation

Later cash can be bigger in nominal terms yet still contribute less to present value if discounting is strong.

System note

Forecast quality, timing assumptions, and discount assumptions all matter. Discounting is only one layer of the problem.

Worked Calculation — Cumulative Risk Exposure

Difficulty: Intermediate

Setup

Suppose a strategy runs absolute leverage levels 0.4, 0.9, 0.6, 0.3 across four equal-length intervals.

Step 1

Gross exposure-time is 0.4 + 0.9 + 0.6 + 0.3 = 2.2 leverage-interval units.

Step 2

If each interval is one hour, accumulated gross exposure equals 2.2 leverage-hours.

Interpretation

This differs from peak leverage, which is only 0.9. Peak and accumulated burden answer different control questions.

System note

Accumulated exposure is useful for compliance summaries, risk-budget consumption analysis, and comparing strategies with different holding durations.

REVIEW / APPENDIX

Diagnostic Tables and Failure Modes

Quant Research Field Guide

Difficulty: Intermediate

Failure-Mode Table

Integral result looks far too large	Units or step size may be wrong. Check whether you multiplied by Δt or used annualized numbers without converting horizon.
Expected value changes when grid spacing changes	Density normalization or numerical integration is unstable. Recheck mass totals and refine the grid.
Backtest PnL looks better gross than net	Execution cost accumulation is missing or understated.
Tail-risk estimate feels too calm	The CDF / density model may underestimate extreme states or use an over-smoothed empirical estimate.
Monte Carlo estimate jumps around	Too few paths or high payoff variance. Increase paths or use variance-reduction techniques.
Conditional expectation is unstable	Feature buckets are sparse or regime classification is noisy.

Diagnostic Questions

What is the integrand?	State what local quantity is being accumulated.
What are the bounds?	Specify the horizon or state region of interest.
What are the units?	Confirm the output units by multiplying units of integrand by units of integration variable.
How is probability represented?	Discrete weights, continuous density, or simulated scenarios.
How sensitive is the result?	Check grid resolution, path count, model assumptions, and tail behavior.

Comparison Sheet — Integrals, Expectations, and Related Obj

Difficulty: Intermediate

Comparison Table

Definite integral	Total accumulation over a bounded interval.
Antiderivative	Function whose derivative recovers the integrand.
Running accumulation function	$A(t)=\int_0^t f(s)ds$; one total for each t.
Discrete expectation	Probability-weighted sum across finitely many states.
Continuous expectation	Probability-weighted integral across a continuum of outcomes.
Monte Carlo expectation	Sample average approximation of an expectation.
Present value	Discounted accumulation of future cash flows.
Expected shortfall	Conditional expectation of tail losses beyond a VaR threshold.
Expected log growth	Growth-oriented objective focused on repeated compounding.
Most likely outcome / mode	Highest local probability density, not necessarily the average.

Formula Summary Sheet

Difficulty: Intermediate

Core Formulas 1

$\int_a^b f(x)dx$; $A(t)=\int_0^t f(s)ds$; $F'(x)=f(x)$; $\int_a^b f(x)dx = F(b)-F(a)$

Integration basics.

Core Formulas 2

$E[X]=\Sigma x_i p_i$; $E[X]=\int x f(x)dx$; $F(x)=\int_{-\infty}^{x} f(u)du$

Expectation and cumulative probability.

Core Formulas 3

$Var(X)=E[(X-E[X])^2]$; $Cov(X,Y)=E[XY]-E[X]E[Y]$

Dispersion and co-movement.

Core Formulas 4

$PV=\int CF(t)D(t)dt$; $Price=e^{-rT}\int payoff(S_T)f_Q(S_T)dS_T$

Valuation and pricing.

Core Formulas 5

$W(t)=W_0 \exp(\int_0^t r(s)ds)$; Growth objective: maximize $E[\log(1+fR)]$

Compounding and growth.

REVIEW / APPENDIX

Quick Recall Review I

Quant Research Field Guide

Difficulty: Intermediate

Quick Questions

- What is the difference between a local rate and an accumulated total?
- Why do log returns add across time while simple returns do not add exactly?
- What is the difference between a density and a probability?
- How would you explain present value as an accumulation problem?
- Why can a trade with frequent small losses still have positive expected value?

Mini Review Prompts

- State the fundamental theorem of calculus in words.
- Write a discrete expectation formula and a continuous expectation formula.
- Explain what expected shortfall adds beyond VaR.
- Give one example of accumulation over time and one example of accumulation over states.

REVIEW / APPENDIX

Quick Recall Review II

Difficulty: Intermediate

Implementation Prompts

- When would you prefer Monte Carlo over a grid integral?
- How would you numerically check whether a density is normalized?
- What diagnostics would you use if an integral estimate changes too much with grid resolution?
- Why should a backtest track gross and net PnL separately?
- How can accumulated exposure differ from peak exposure in interpretation?

Common Confusions to Rehearse

- Expected value vs median vs mode.
- Compounding vs discounting.
- Definite integral vs antiderivative.
- Probability density vs probability mass.
- Expected return vs expected log growth.

REVIEW / APPENDIX

Final Review and Concept Map

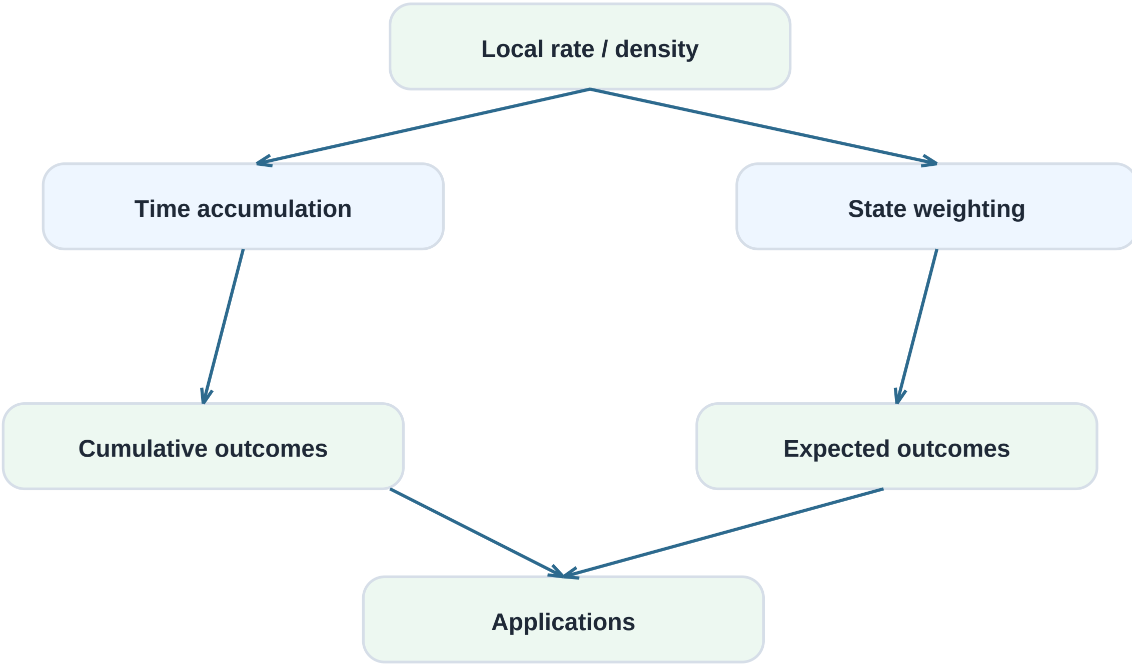
Quant Research Field Guide

Difficulty: Intermediate

Big Picture Summary

This manual revolves around one master idea: local contributions accumulate into meaningful totals. Over time, that creates cumulative return, cost, exposure, and present value. Across states, that creates expected value, option value, and tail-risk summaries. The research skill is learning which local quantity to model, how to weight it, and how to aggregate it reliably.

Concept Map



Applications include compounding, discounted cash flow, PnL, scenario analysis, option valuation, and tail risk.

Related-Term Index

Difficulty: Intermediate

Index Notes

Use the index as a final memory jogger. If you can explain each term in a sentence and say where it appears in a research workflow, the core learning objective of this manual has been met.

Alphabetical Term List

- Accumulation function, Antiderivative, Area under the curve, CDF, Change of variables, Conditional expectation, Continuous compounding, Covariance, Density, Definite integral, Discount factor, Double integral, Expected payoff, Expected shortfall, Expected value, Fundamental theorem of calculus, Geometric growth, Integration by parts, Joint density, Kelly logic, Law of iterated expectations, Linearity of expectation, Log return, Mode, Monte Carlo, Numerical integration, Option payoff, Path dependence, Present value, Probability weight, Riemann sum, Scenario analysis, Signed area, Simpson’s rule, Tail probability, Time average, Trapezoid rule, Variance.